



Shaping processes for polymers

Groover M.P.: Chapter 10



Plastics can be shaped into a wide variety of products:

- Molded parts
- Extruded sections
- Films
- Sheets
- Insulation coatings on electrical wires
- Fibers for textiles

In addition, plastics are often the principal ingredient in other materials, such as:

- Paints and varnishes
- Adhesives
- Various polymer matrix composites

Many plastic shaping processes can be adapted to produce items made of rubbers and polymer matrix composites.



PLASTIC SHAPING PROCESSES ARE IMPORTANT

Almost unlimited variety of part geometries.

Plastic molding is a *net shape* process:

- Further shaping is not needed.

Less energy is required than for metals due to much lower processing temperatures:

- Handling of product is simplified during production because of lower temperatures.

Painting or plating is usually not required.



To shape a thermoplastic polymer it must be heated so that it softens to the consistency of a liquid.

In this form, it is called a *polymer melt*.

Important properties of polymer melts:

- Viscosity
- Viscoelasticity

Due to its high molecular weight, a polymer melt is a thick fluid with high viscosity.

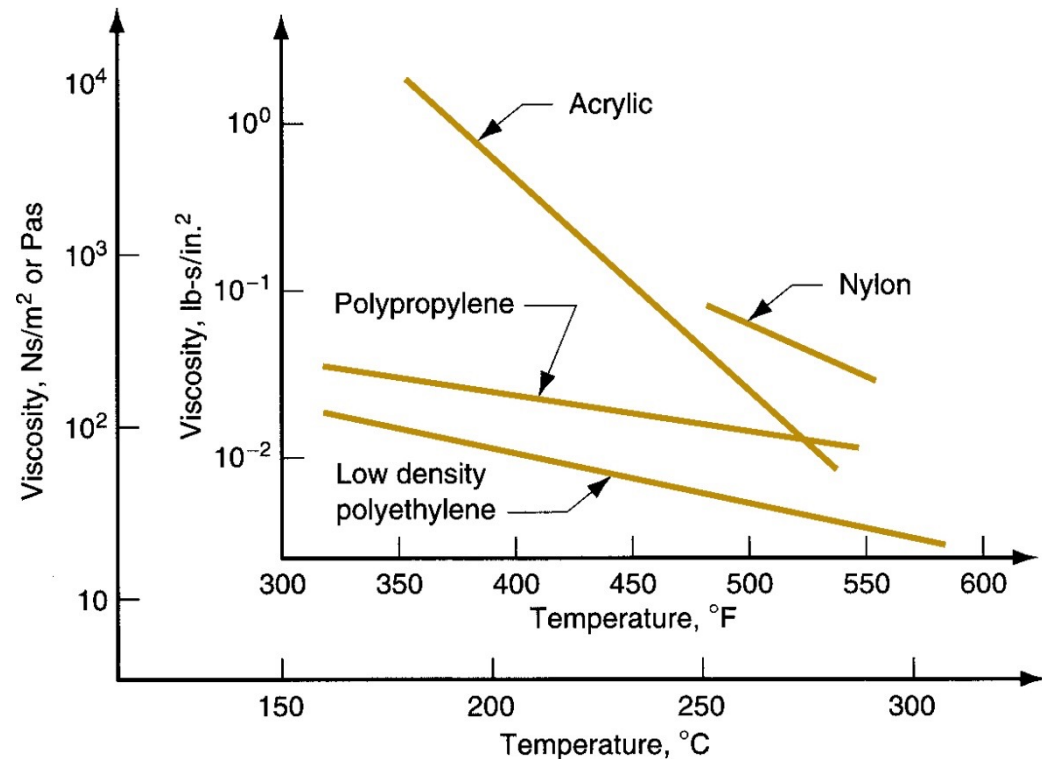
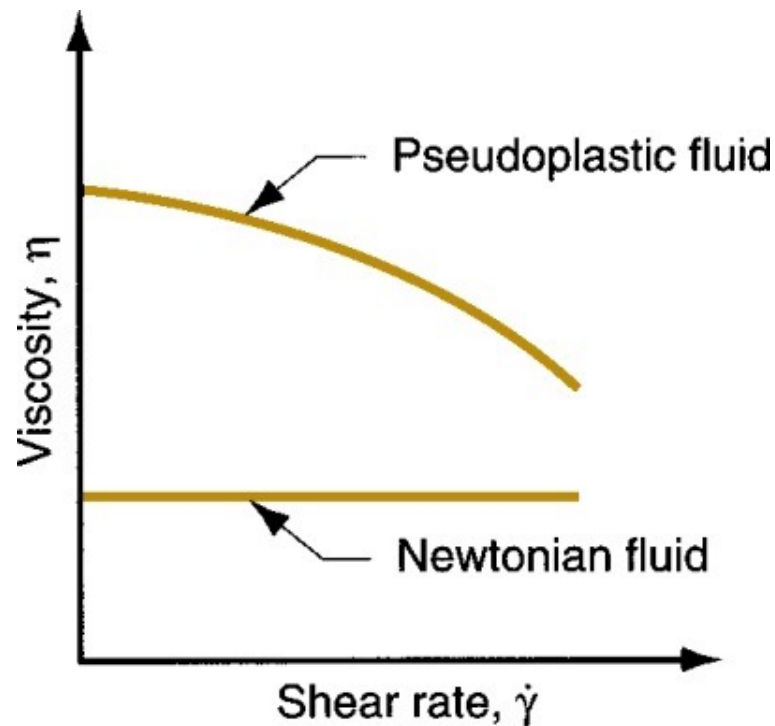
Most polymer shaping processes involve flow through small channels or die openings. Flow rates are often large, leading to high shear rates and shear stresses, so significant pressures are required in these processes.



VISCOSITY, SHEAR RATE AND TEMPERATURE

Viscosity of a polymer melt decreases with either shear rate or temperature.

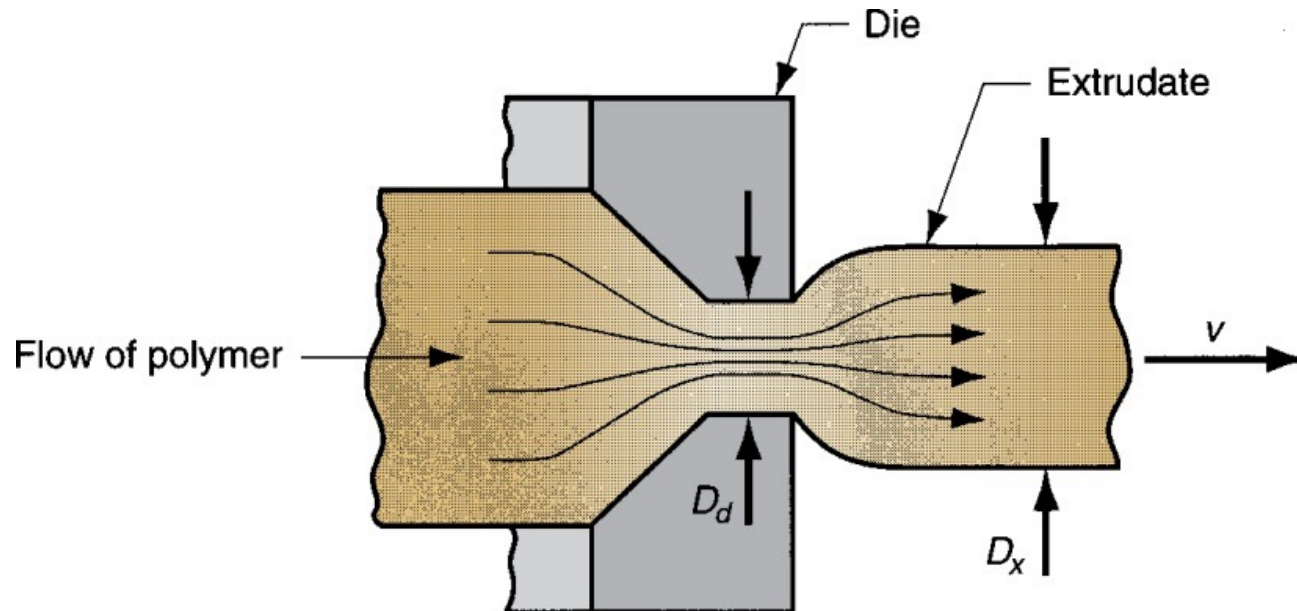
Thus the fluid becomes thinner (flows more easily) at either higher shear rates or higher temperatures.





Combination of viscosity and elasticity

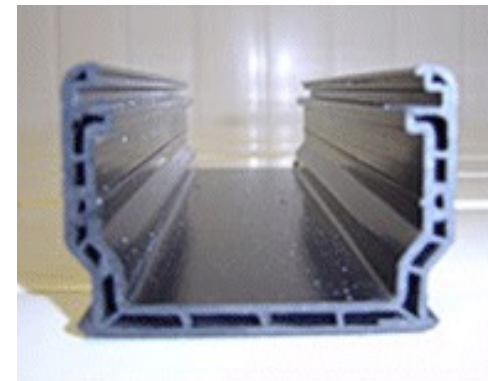
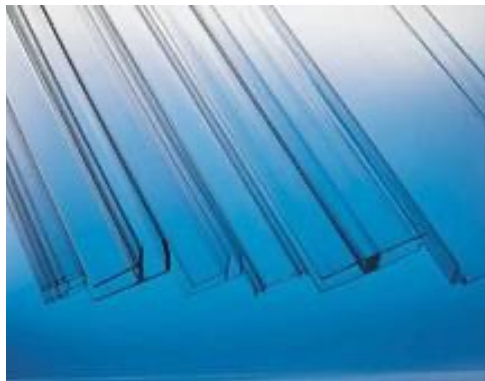
Example: *die swell* in extrusion - hot plastic expands when exiting the die opening.





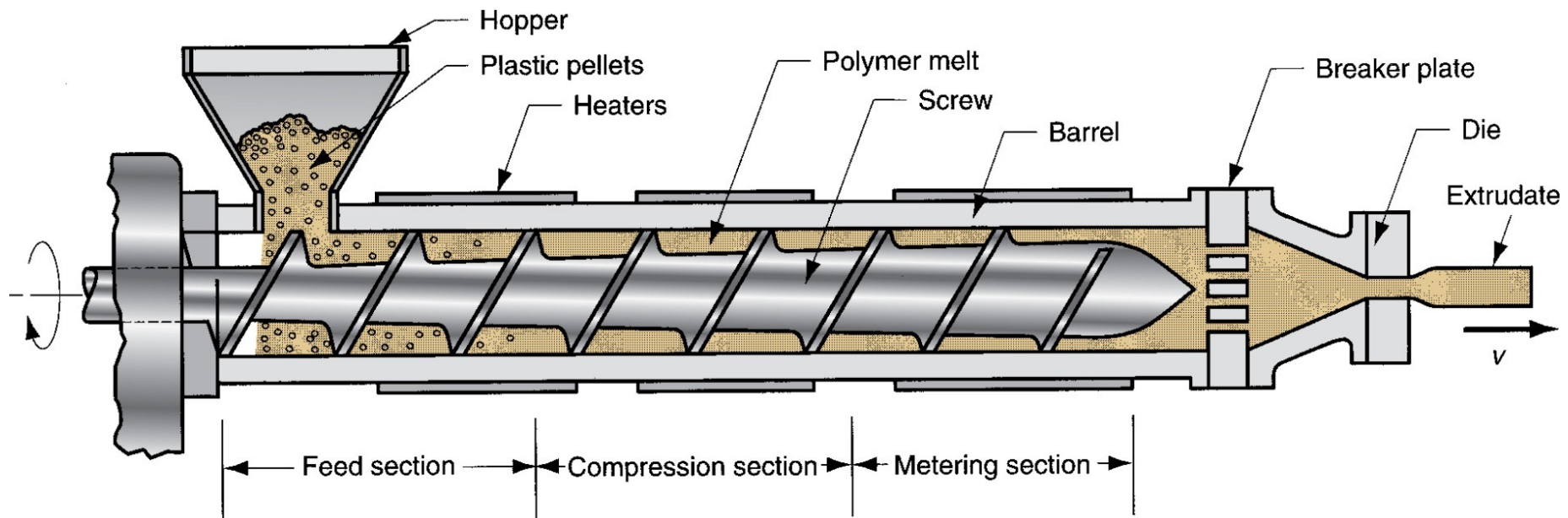
Material is forced to flow through a die orifice to provide long continuous product whose cross section is determined by the shape of the orifice.

- Widely used for thermoplastics and elastomers to mass produce items such as tubing, pipes, hose, structural shapes, sheet and film, continuous filaments, and coated electrical wire.
- Carried out as a continuous process; *extrudate* is then cut into desired lengths.





Components and features of a (single-screw) extruder for plastics and elastomers.



Two main components of an extruder:

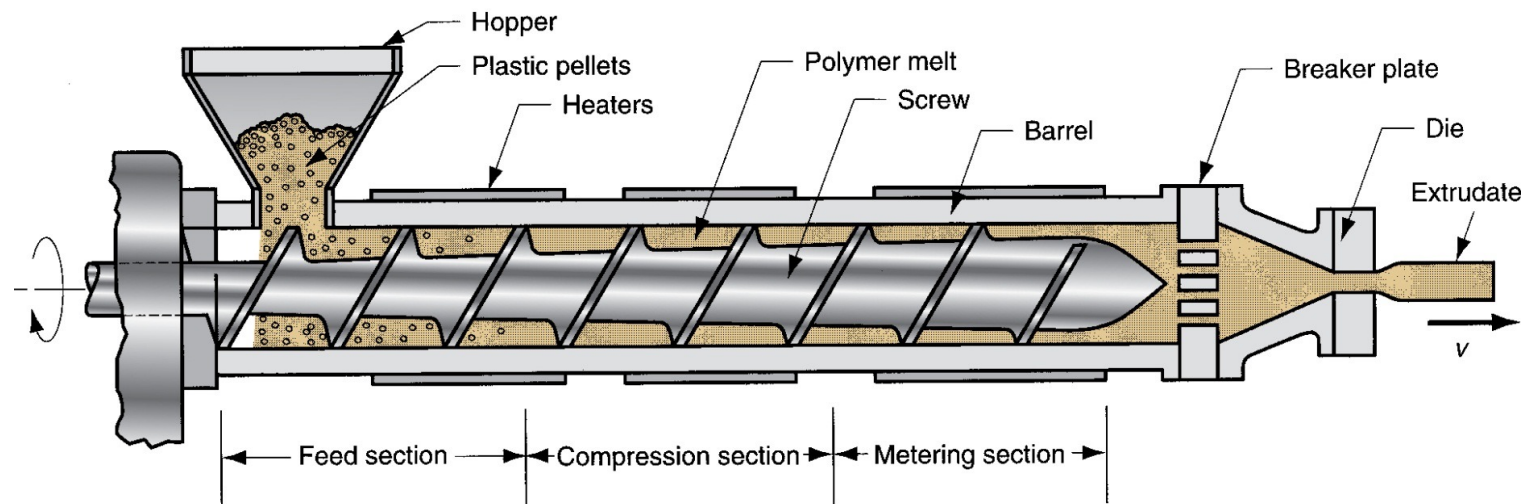
1. Barrel
2. Screw

The die is not an extruder component: It is a special tool that must be fabricated for the profile to be produced.



EXTRUDER BARREL

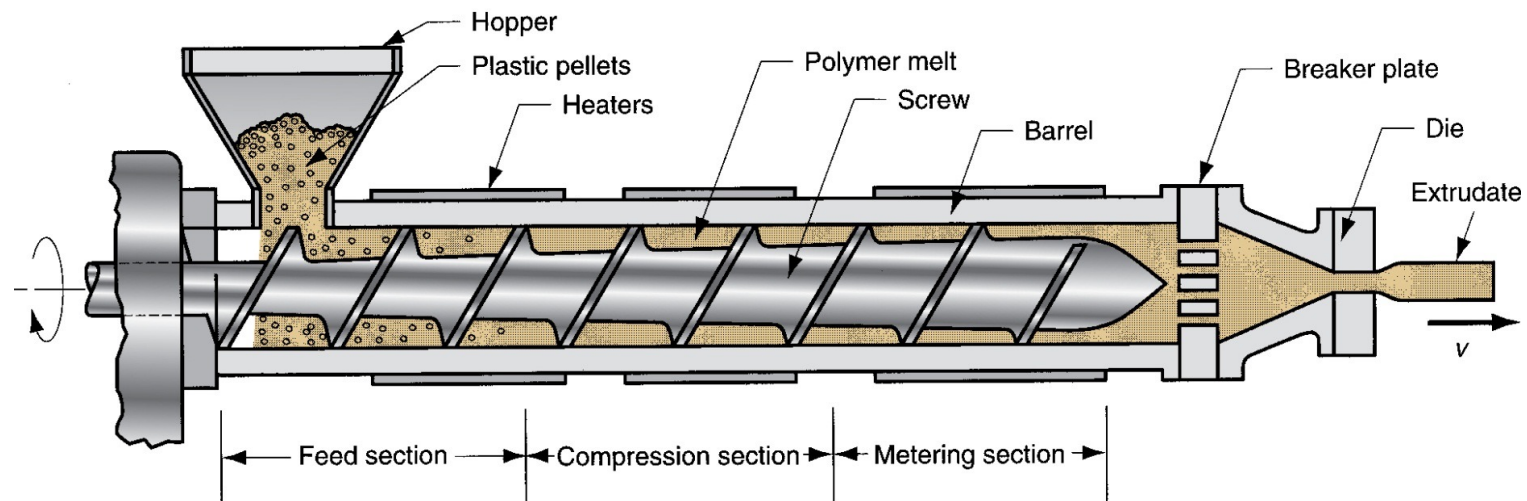
- Internal diameter typically ranges from 25 to 150 mm.
- L/D ratios usually ~ 10 to 30 : higher ratios for thermoplastics, lower ratios for elastomers.
- Feedstock fed by gravity onto screw whose rotation moves material through barrel.
- Electric heaters melt feedstock; subsequent mixing and mechanical working adds heat which maintains the melt.





Divided into sections to serve several functions:

- Feed section - feedstock is moved from hopper and preheated.
- Compression section - polymer is transformed into thick fluid, air mixed with pellets is extracted from melt, and material is compressed.
- Metering section - melt is homogenized and sufficient pressure developed to pump it through die opening.



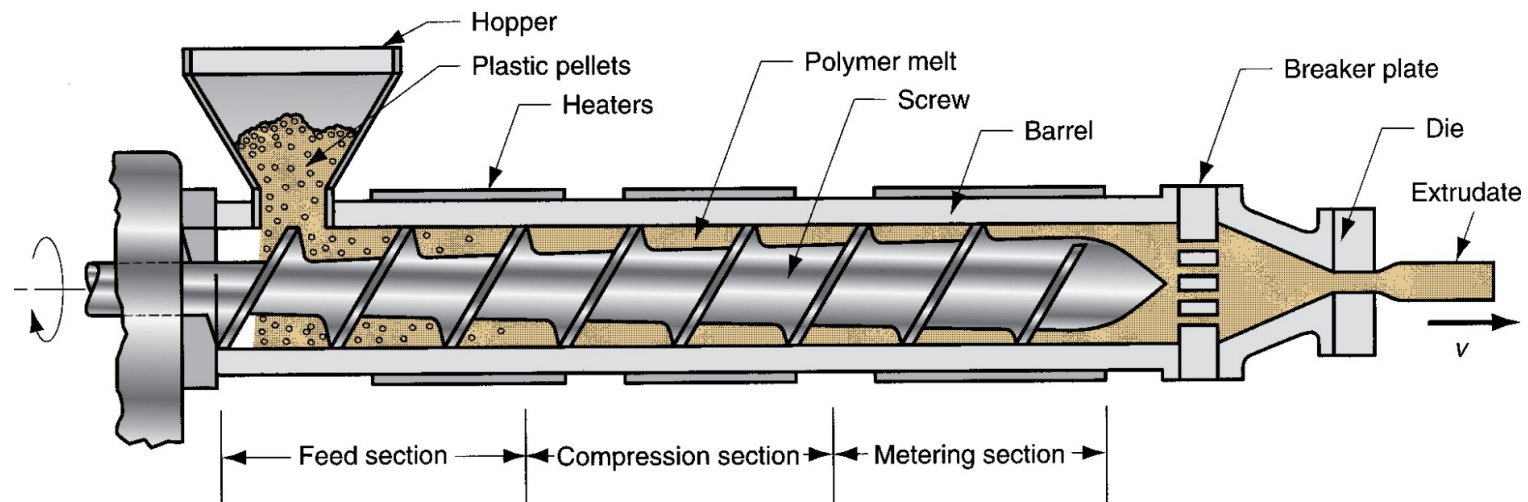


DIE END OF EXTRUDER

Progress of polymer melt through barrel leads ultimately to the die zone. Before reaching the die, the melt passes through a *screen pack* - series of wire meshes supported by a stiff plate containing small axial holes.

Functions of screen pack:

- Filters out contaminants and hard lumps
- Builds pressure in metering section
- Straightens flow of polymer melt and removes its "memory" of circular motion from screw



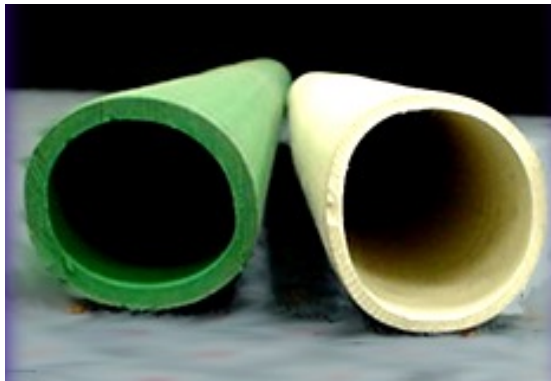


DIE CONFIGURATIONS AND EXTRUDED PRODUCTS

The shape of the die orifice determines the cross-sectional shape of the extrudate.

Common die profiles and corresponding extruded shapes:

- Solid profiles
- Hollow profiles, such as tubes
- Wire and cable coating
- Sheet and film
- Filaments



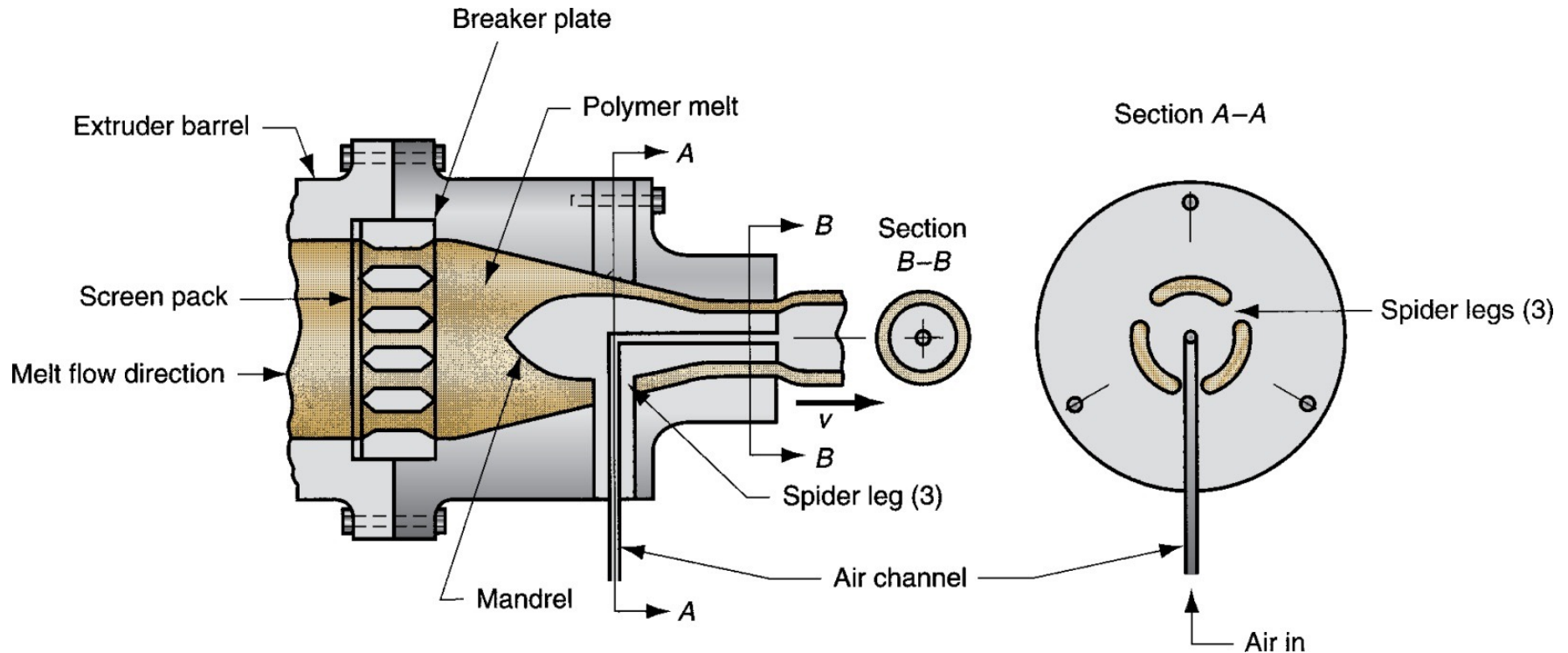


EXTRUSION DIE FOR HOLLOW SHAPES

Side view of extrusion die for hollow cross sections.

Section A-A is a front view of how mandrel is held in place.

Section B-B shows tube prior to exiting die; die swell enlarges diameter.





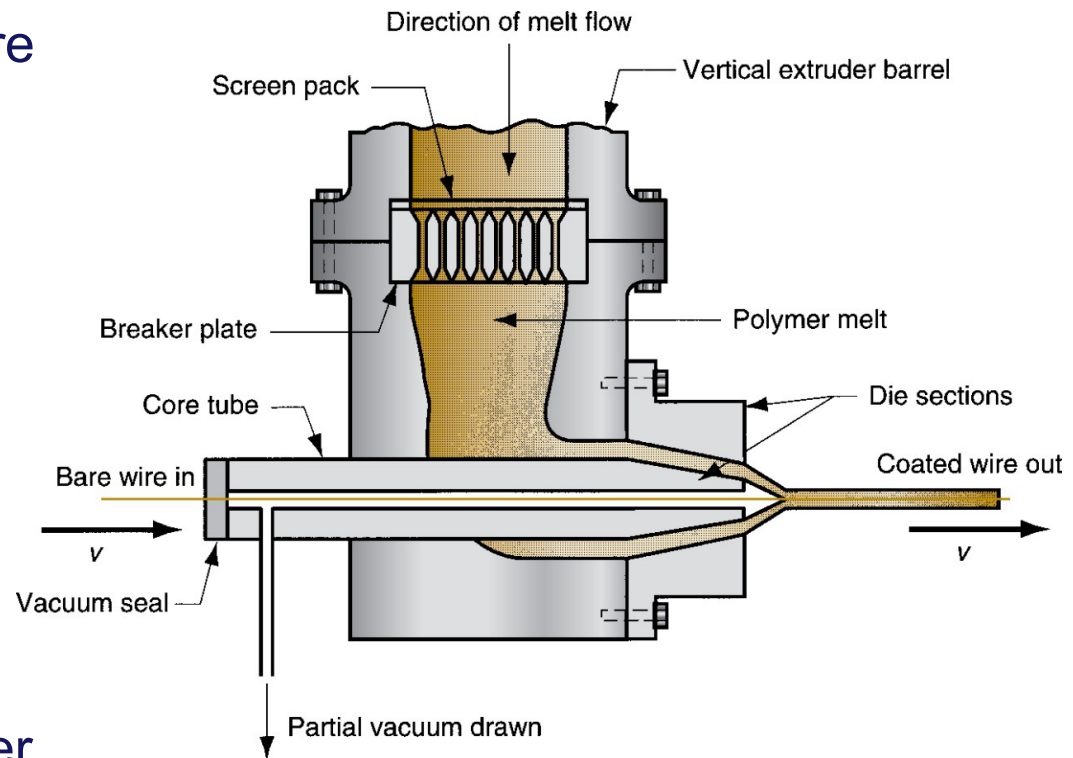
Polymer melt is applied to bare wire as it is pulled at high speed through die:

- A slight vacuum is drawn between wire and polymer to promote adhesion of coating.

Wire provides rigidity during cooling:

- Usually aided by passing coated wire through a water trough.

Product is wound onto large spools at speeds up to 50 m/s.





POLYMER SHEET AND FILM

Film - thickness below 0,5 mm

- Packaging - product wrapping material, grocery bags, and garbage bags
- Stock for photographic film
- Pool covers and irrigation ditch liners

Sheet - thickness from 0,5 mm to about 12,5 mm

- Flat window glazing
- Thermoforming stock

All thermoplastic polymers

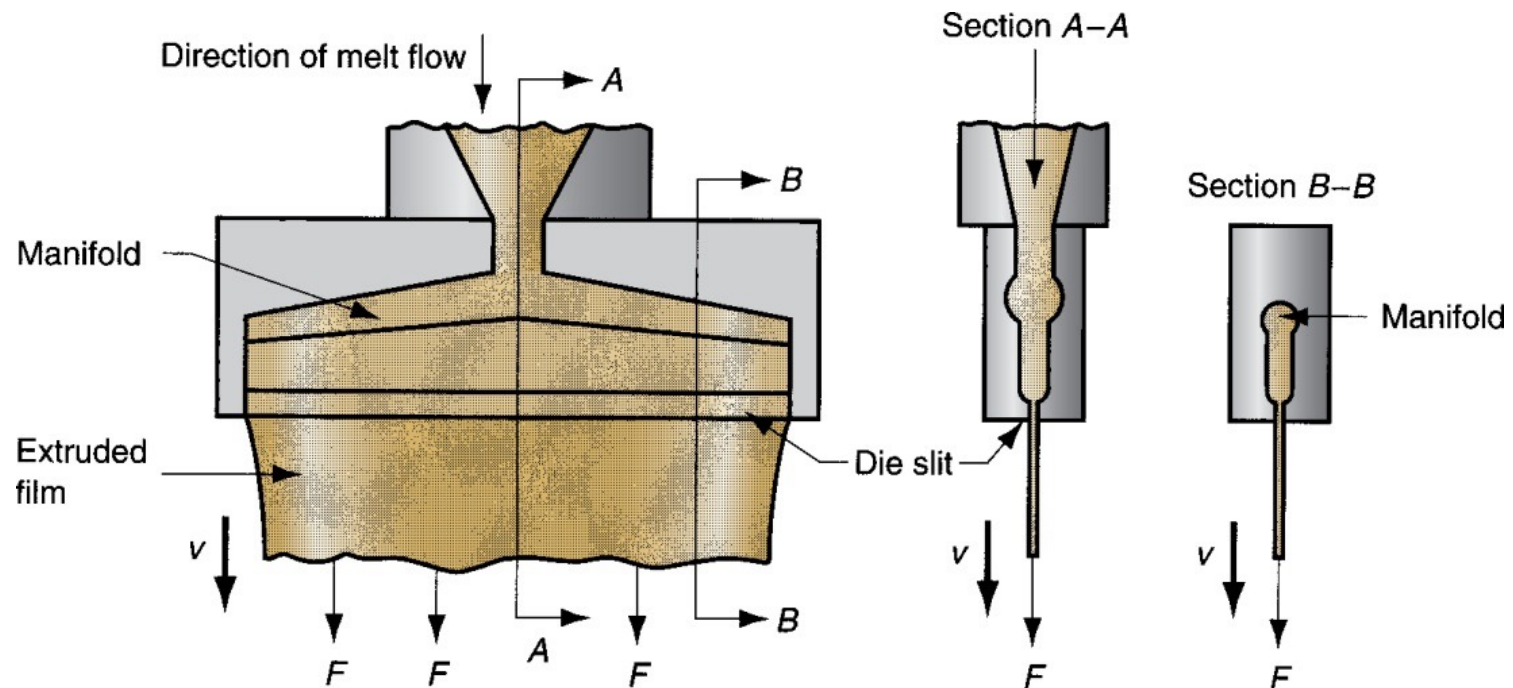
- Polyethylene, mostly low density PE (LDPE)
- Polypropylene (PP)
- Polyvinylchloride (PVC)
- Cellophane



SLIT-DIE EXTRUSION OF SHEET AND FILM

Production of sheet and film by conventional extrusion, using a narrow slit as the die opening:

- Slit may be up to 3 m wide and as narrow as around 0,4 mm
- A problem is thickness uniformity throughout width of stock, due to drastic shape change of polymer melt as it flows through die
- Edges of film usually must be trimmed because of thickening at edges



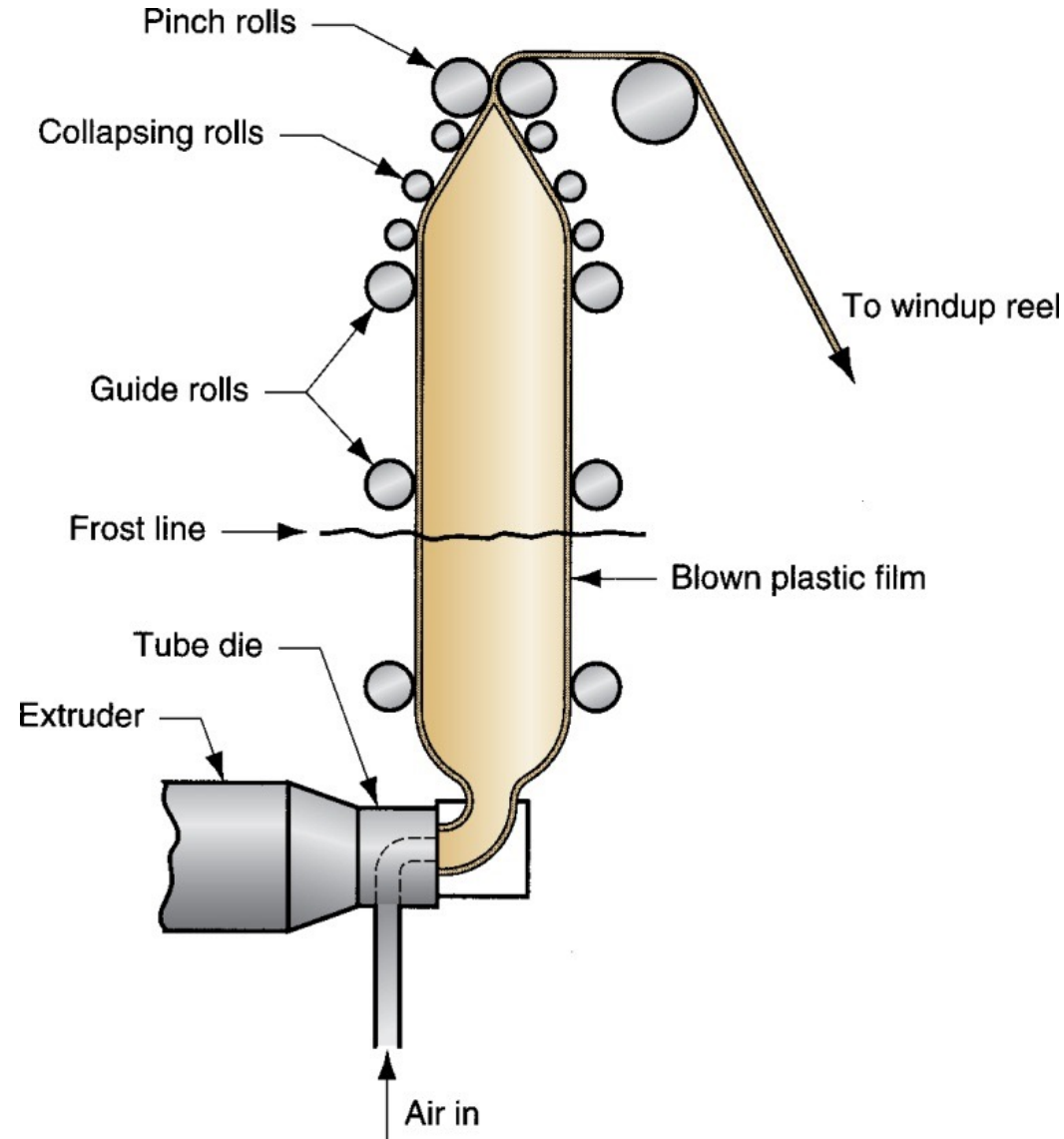


BLOWN-FILM EXTRUSION PROCESS

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Combines extrusion and blowing to produce a tube of thin film

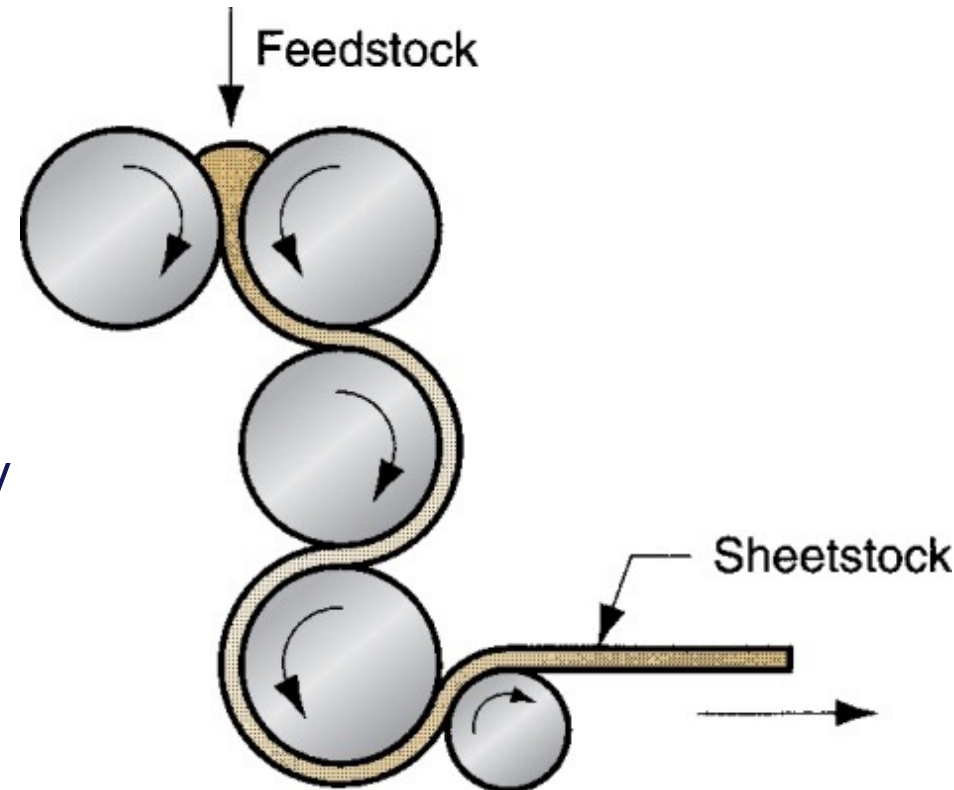
- Process sequence:
 1. Extrusion of tube
 2. Tube is drawn upward while still molten and expanded by air inflated into it through die
- Air is blown into tube to maintain uniform film thickness and tube diameter





Feedstock is passed through a series of rolls to reduce thickness to desired gage:

- Expensive equipment, high production rates.
- Process is noted for good surface finish and high gage accuracy.
- Typical materials: rubber or rubbery thermoplastics such as plasticized PVC.
- Products: PVC floor covering, shower curtains, vinyl tablecloths, pool liners, and inflatable toys.





FIBER AND FILAMENT PRODUCTS

Definitions:

- *Fiber* - a long, thin strand whose length is at least 100 times its cross-section
- *Filament* - a fiber of continuous length

Applications:

- Fibers and filaments for textiles
- Reinforcing materials in polymer composites

Fibers can be natural or synthetic:

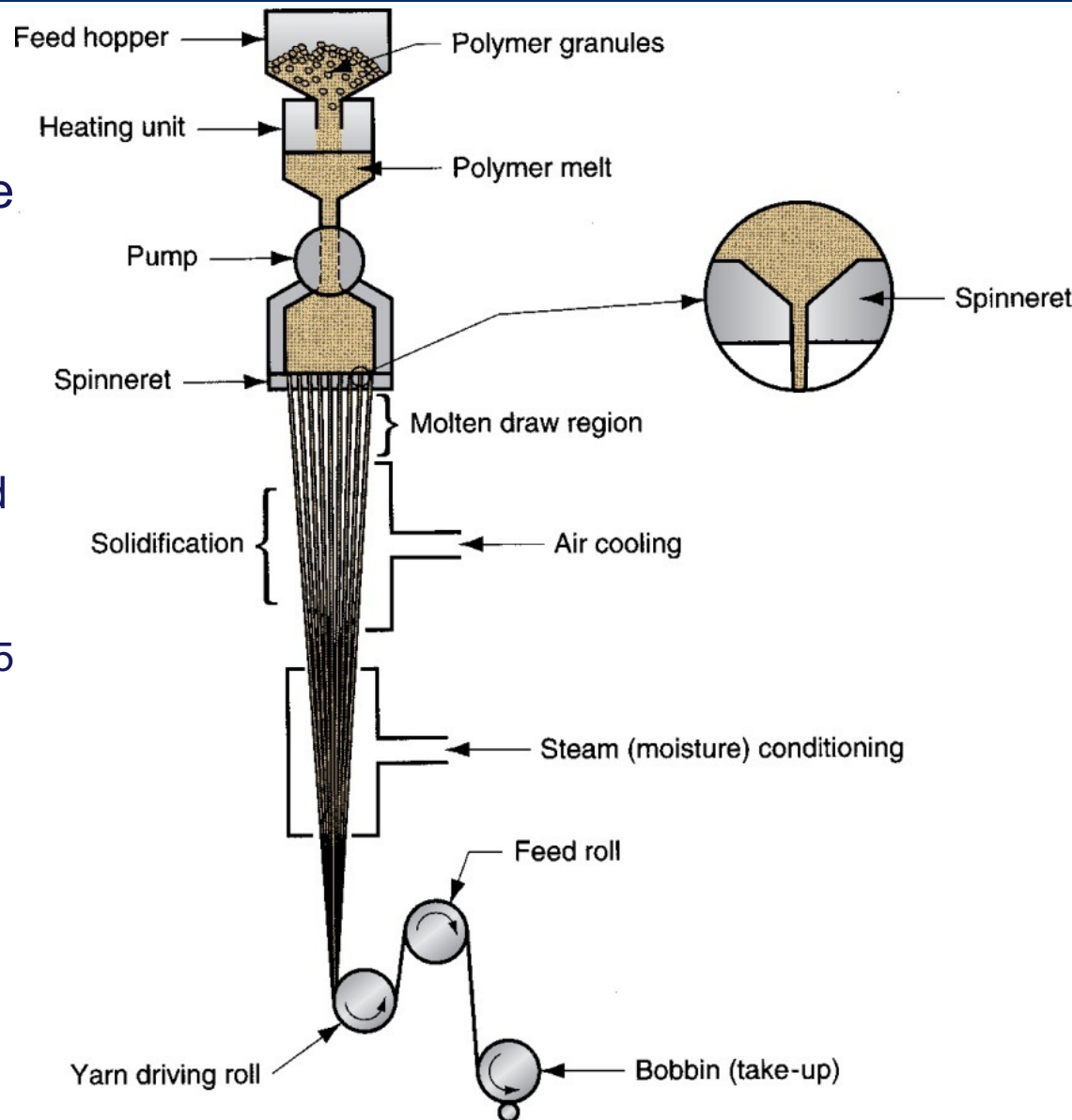
- Natural fibers constitute ~ 25% of total market
 - Cotton is by far the most important staple
 - Wool production is much less than cotton
- Synthetic fibers constitute ~ 75% of total fiber market
 - Polyester is the most important
 - Others: nylon, acrylics, and rayon



For synthetic fibers, *spinning* = extrusion of polymer melt or solution through a *spinneret* (die with multiple small holes), then drawing and winding onto a *bobbin*.

Melt Spinning: Starting polymer is heated to molten state and pumped through spinneret.

- Typical spinneret is 6 mm thick and contains ~ 50 holes of diameter 0,25 mm.
- Filaments are drawn and air cooled before being spooled onto bobbin.
- Final diameter wound onto bobbin may be only 1/10 of extruded size.
- Used for polyester and nylon filaments.





INJECTION MOLDING

Polymer is heated to a highly plastic state and forced to flow under high pressure into a mold cavity where it solidifies, and the *molding* is then removed from the cavity.

- Produces discrete components to net shape.
- Typical cycle time ~10 to 30 s, but cycles of one minute or more are not uncommon.
- Mold may contain multiple cavities, so multiple moldings are produced each cycle.

Injection molding is the most widely used molding process for thermoplastics.

Some thermosets and elastomers are injection molded:

- Modifications in equipment and operating parameters must be made to avoid premature cross-linking of these materials before injection.



INJECTION MOLDED PARTS

Complex and intricate shapes are possible.

Shape limitations:

- Capability to fabricate a mold whose cavity is the same geometry as part;
- Shape must allow for part removal from mold.

Part size from ~ 50 g up to ~ 25 kg.

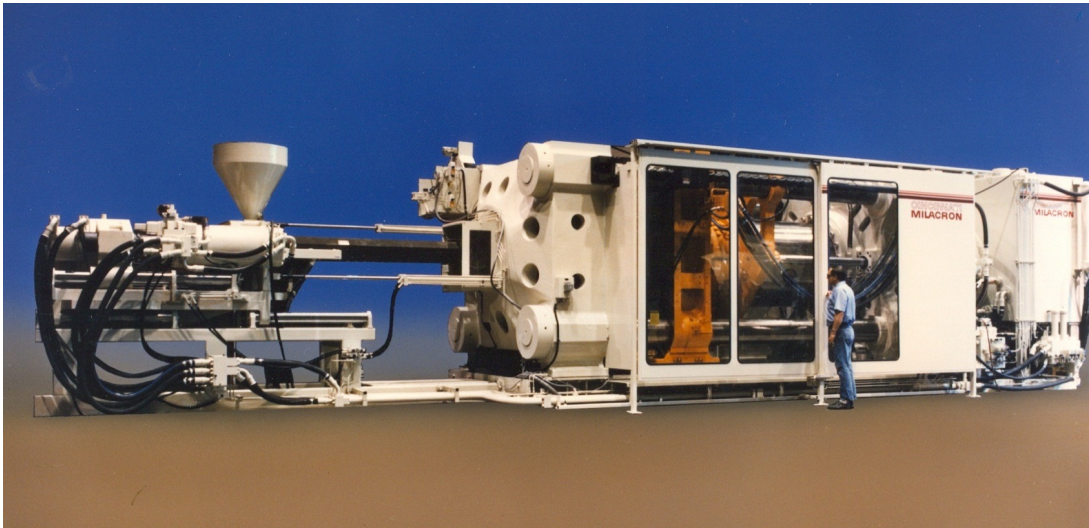
Injection molding is economical only for large production quantities due to high cost of mold.



INJECTION MOLDING MACHINE

Two principal components:

1. Injection unit
 - Melts and delivers polymer melt
 - Operates much like an extruder
2. Clamping unit
 - Opens and closes mold each injection cycle

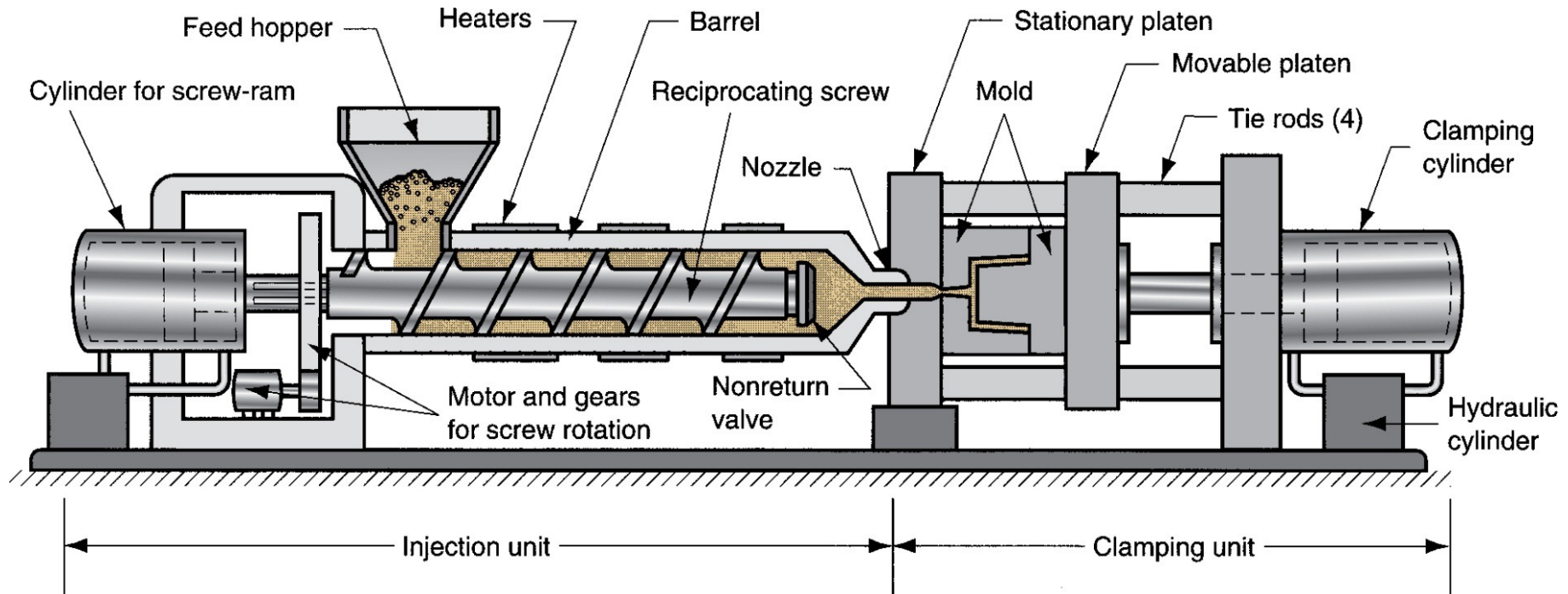


Several clamping designs

- Mechanical;
- Hydraulic.



INJECTION MOLDING MACHINE

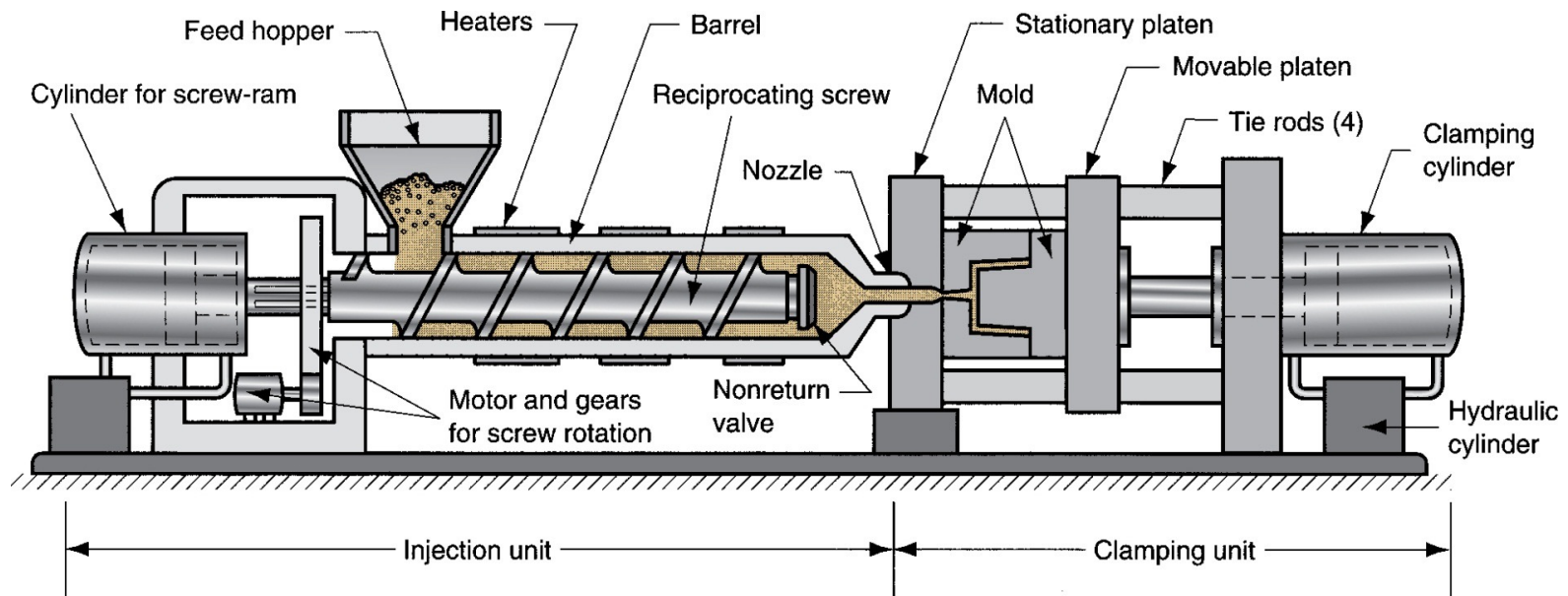




INJECTION UNIT OF MOLDING MACHINE

Consists of *barrel* fed from one end by hopper containing supply of plastic pellets. Inside the barrel is a screw with two functions:

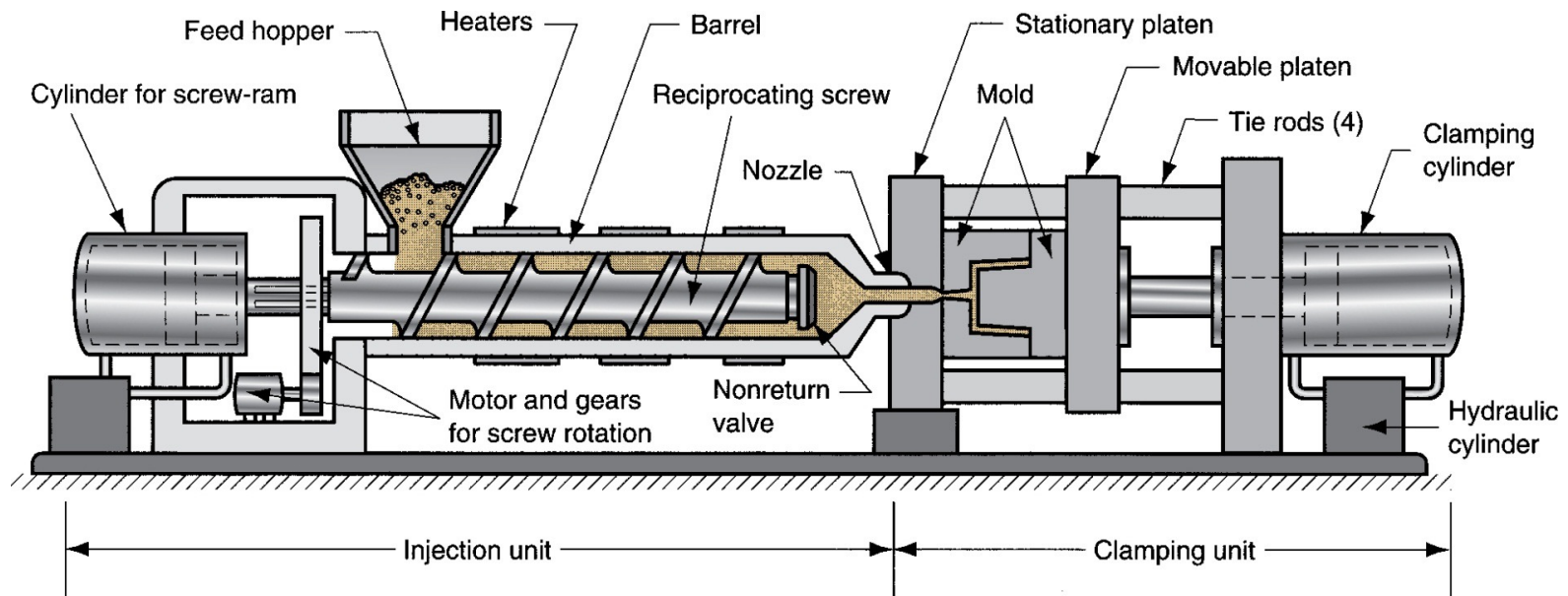
1. Rotates for mixing and heating polymer
2. Acts as a ram (i.e., plunger) to inject molten plastic into mold
 - Non-return valve near tip of screw prevents melt from flowing backward along screw threads
 - Later in cycle ram retracts to its former position



CLAMPING UNIT OF MOLDING MACHINE

Functions:

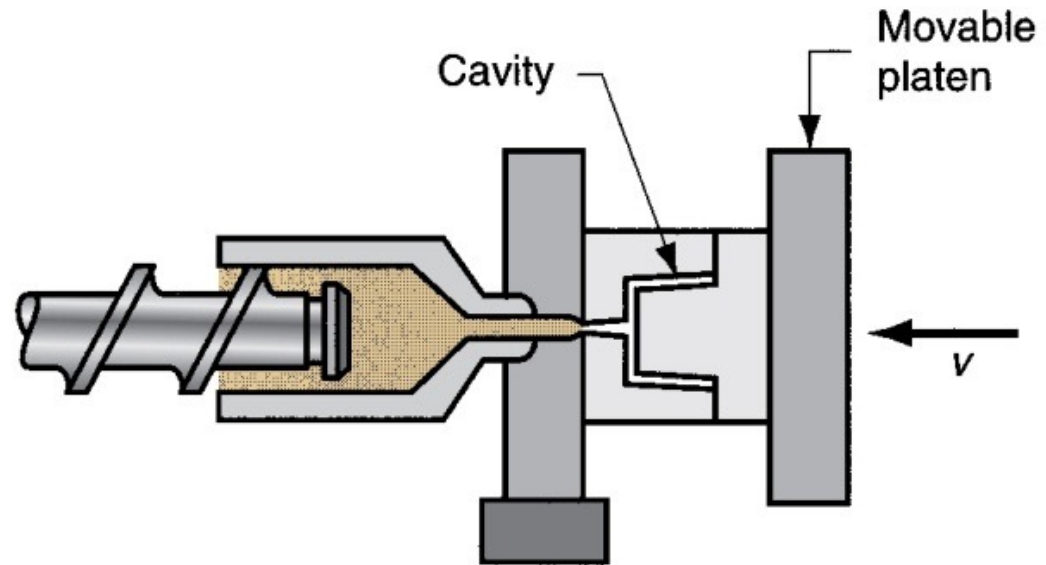
1. Holds two halves of mold in proper alignment with each other;
2. Keeps mold closed during injection by applying a clamping force sufficient to resist injection force;
3. Opens and closes mold at the appropriate times in molding cycle.



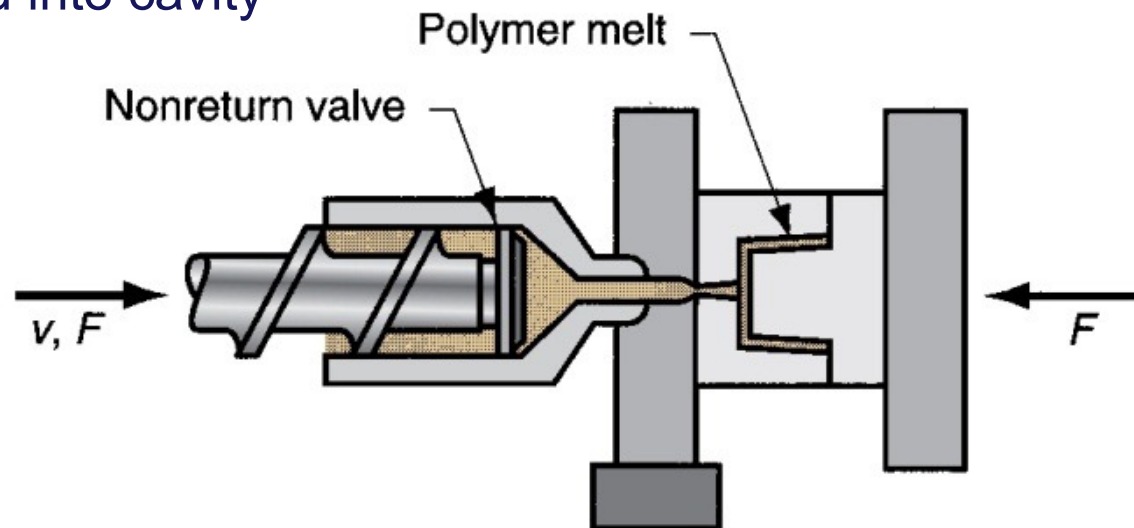


INJECTION MOLDING CYCLE

(1) Mold is closed

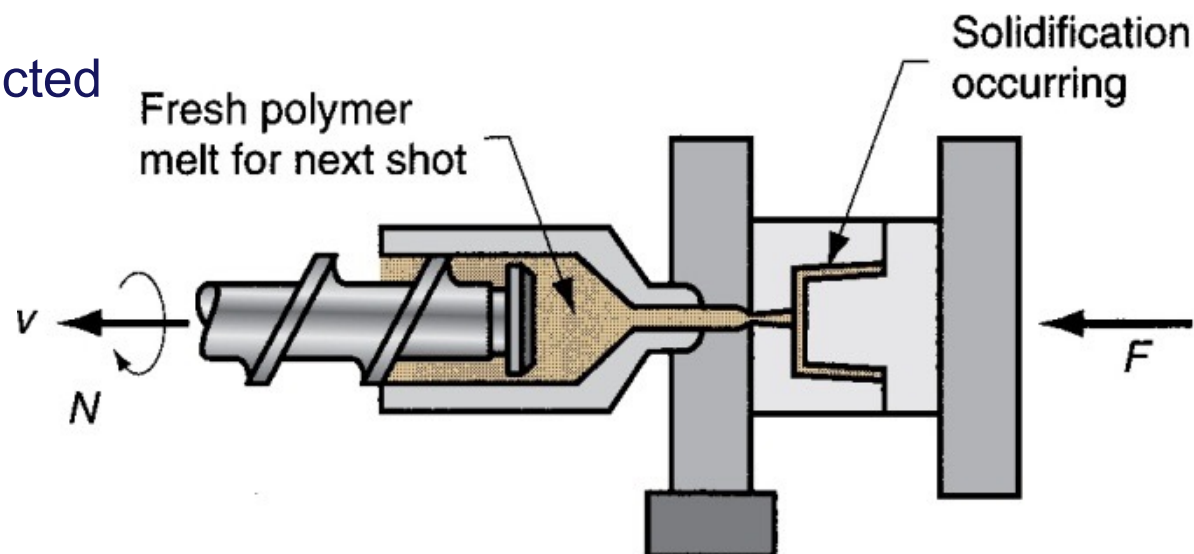


(2) Melt is injected into cavity

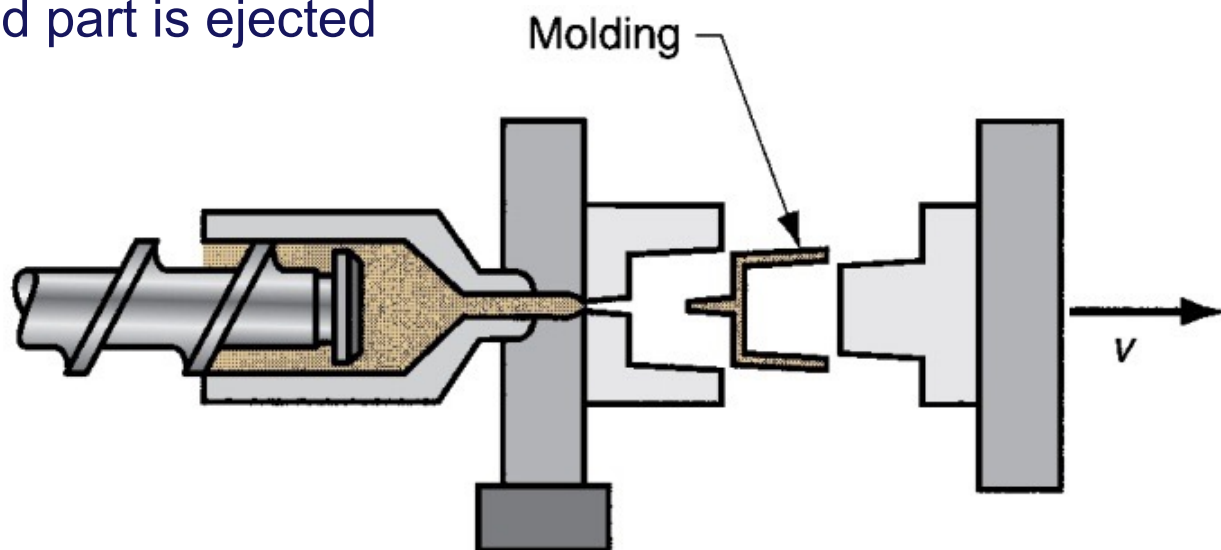




3) screw is retracted



(4) mold is opened, and part is ejected







Polymers have high thermal expansion coefficients, so significant shrinkage occurs during solidification and cooling in mold.

<i>Plastic</i>	<i>Typical Shrinkage, mm/mm</i>
Nylon-6,6	0,020
Polyethylene	0,025
Polystyrene	0,004
PVC	0,005

Dimensions of mold cavity must be larger than specified part dimensions:

$$D_c = D_p + D_p S + D_p S^2$$

Where:

- D_c = dimension of cavity;
- D_p = molded part dimension;
- S = shrinkage value.

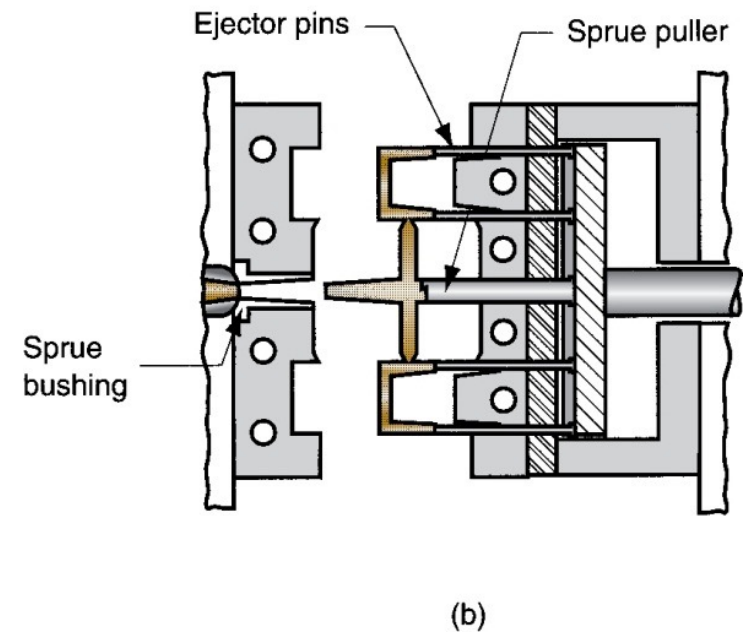
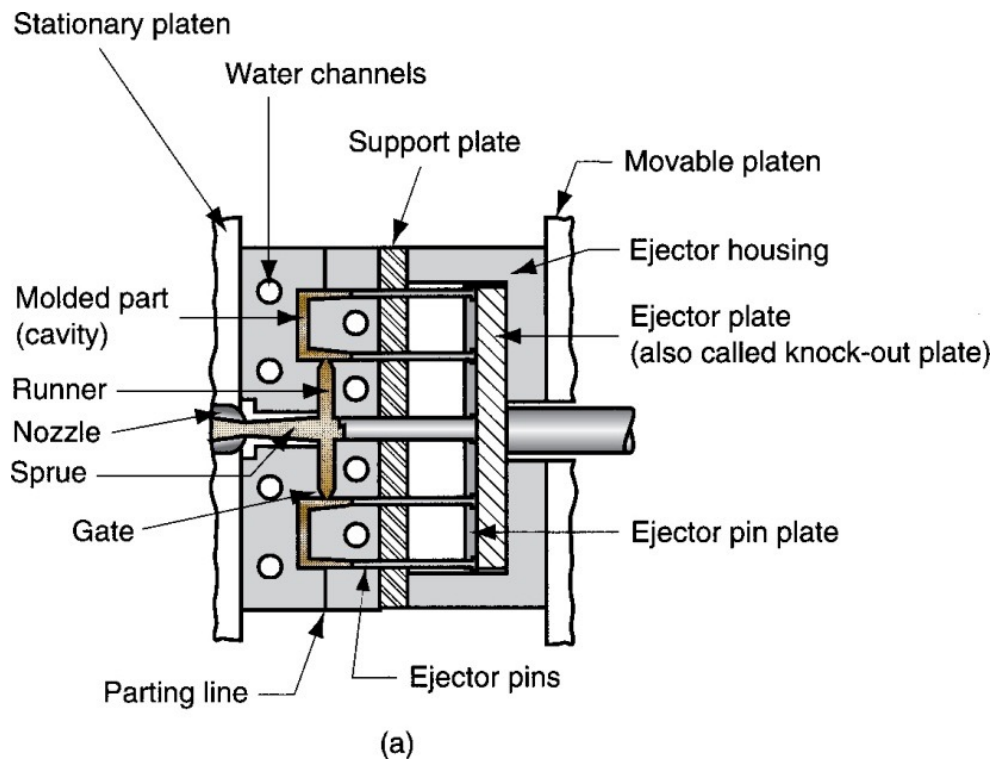
Third term on right hand side corrects for shrinkage in the shrinkage

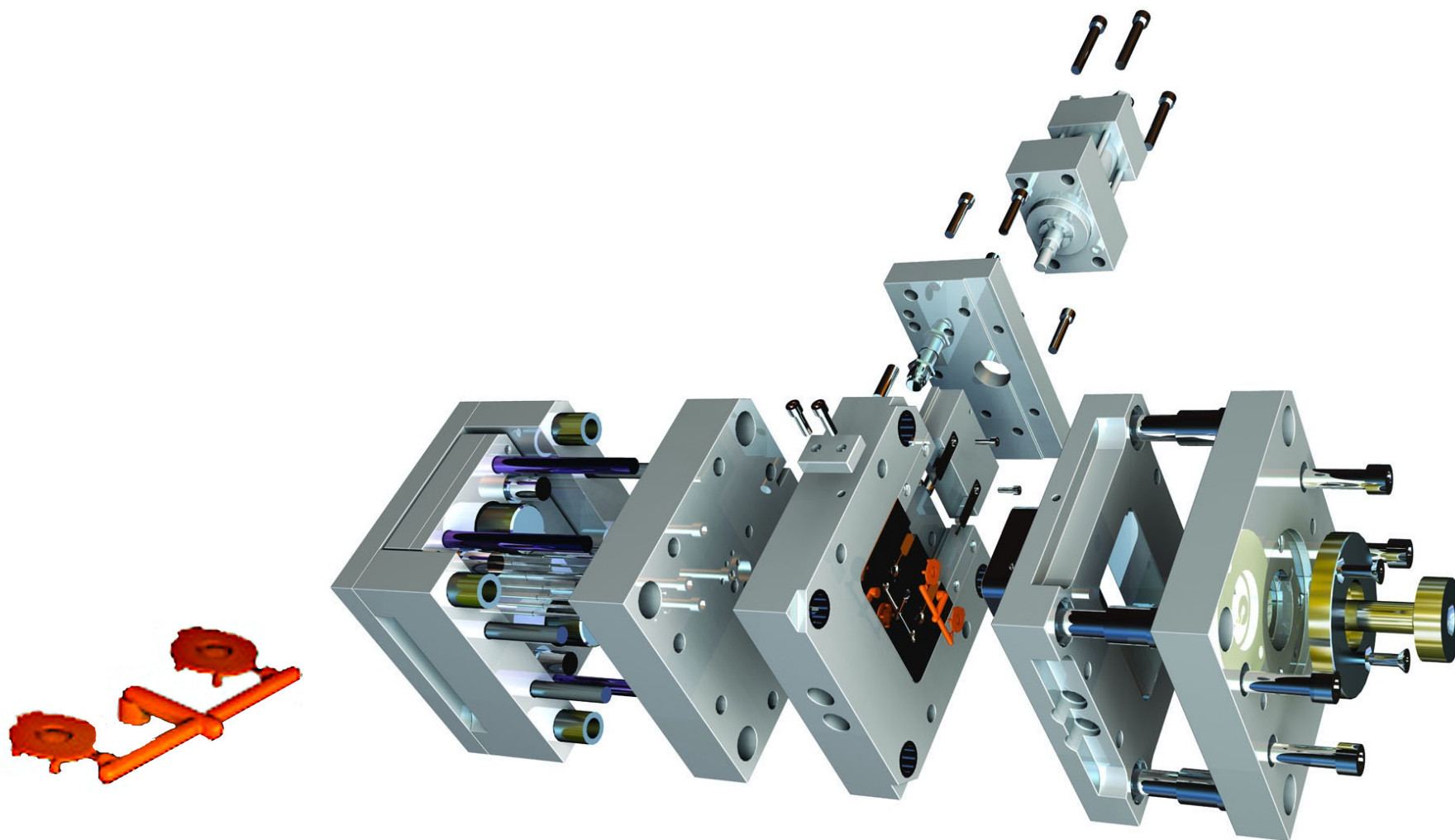


SHRINKAGE FACTORS

- Fillers in the plastic tend to reduce shrinkage.
- Injection pressure – higher pressures force more material into mold cavity to reduce shrinkage.
- Compaction time – similar effect – longer time forces more material into cavity to reduce shrinkage.
- Molding temperature - higher temperatures lower polymer melt viscosity, allowing more material to be packed into mold to reduce shrinkage.

The special tooling in injection molding – custom designed and fabricated for the part to be produced.

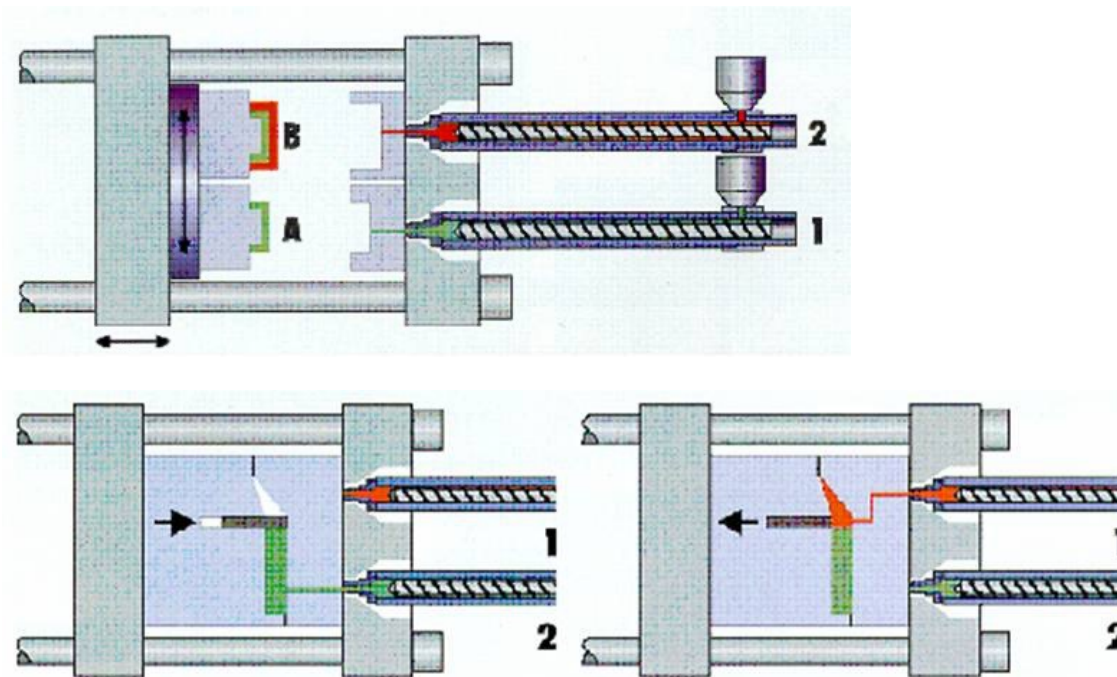






MULTICOMPONENT INJECTION MOLDING

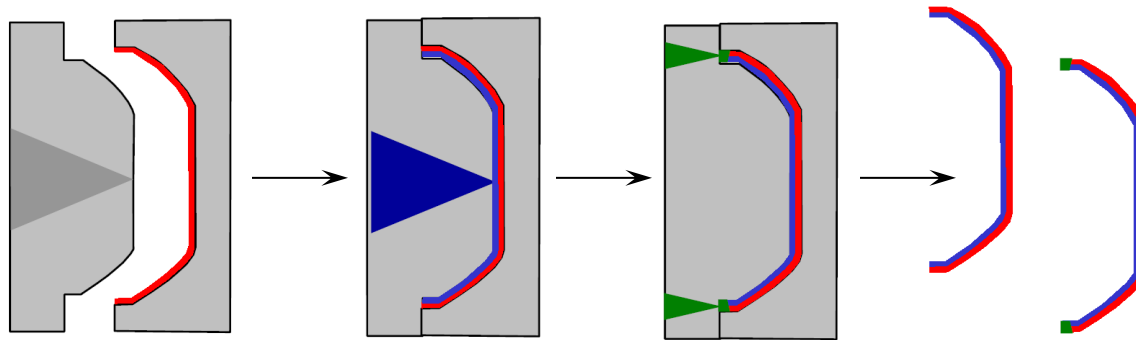
Co-Injection or Sandwich Molding





INJECTION MOLDING AND ASSEMBLING

Overmolding





INJECTION MOLDING OF THERMOSETS

Equipment and operating procedure must be modified to avoid premature cross-linking of TS polymer:

- Reciprocating-screw injection unit with shorter barrel length;
- Temperatures in barrel are relatively low.

Melt is injected into a heated mold, where cross-linking occurs to cure the plastic:

- Curing is the most time-consuming step in cycle;
- Mold is then opened, and part is removed.



ROTATIONAL MOLDING

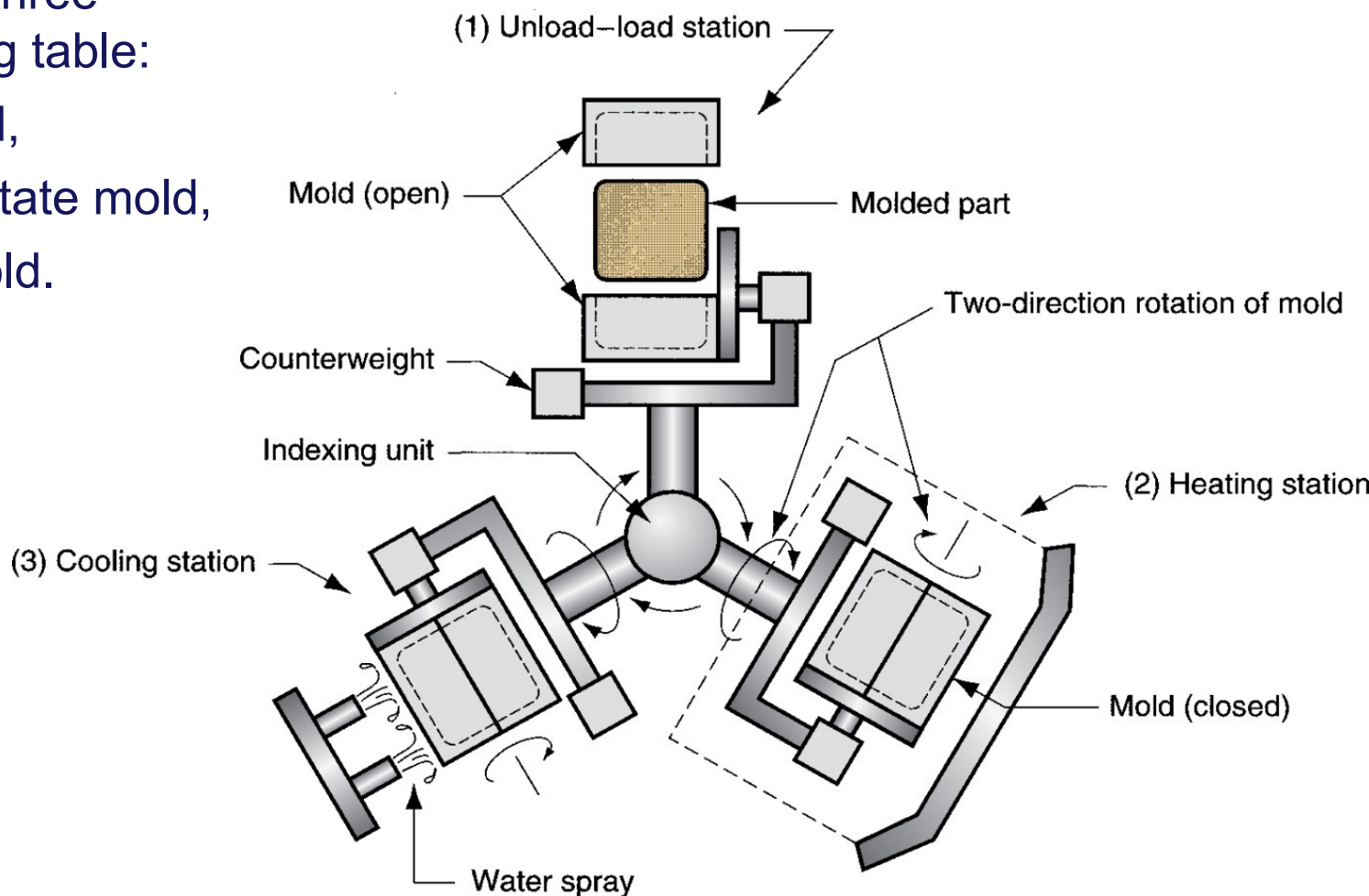
Process uses gravity inside a rotating mold to produce a hollow part shape:

- Alternative to blow molding for larger parts with more complex external geometries
 - But lower production rates than blow molding
- Mostly for thermoplastic polymers but thermoset and elastomer applications are not uncommon
- Products: boat hulls, sandboxes, buoys, garbage cans, large containers and storage tanks



Performed on three-station indexing table:

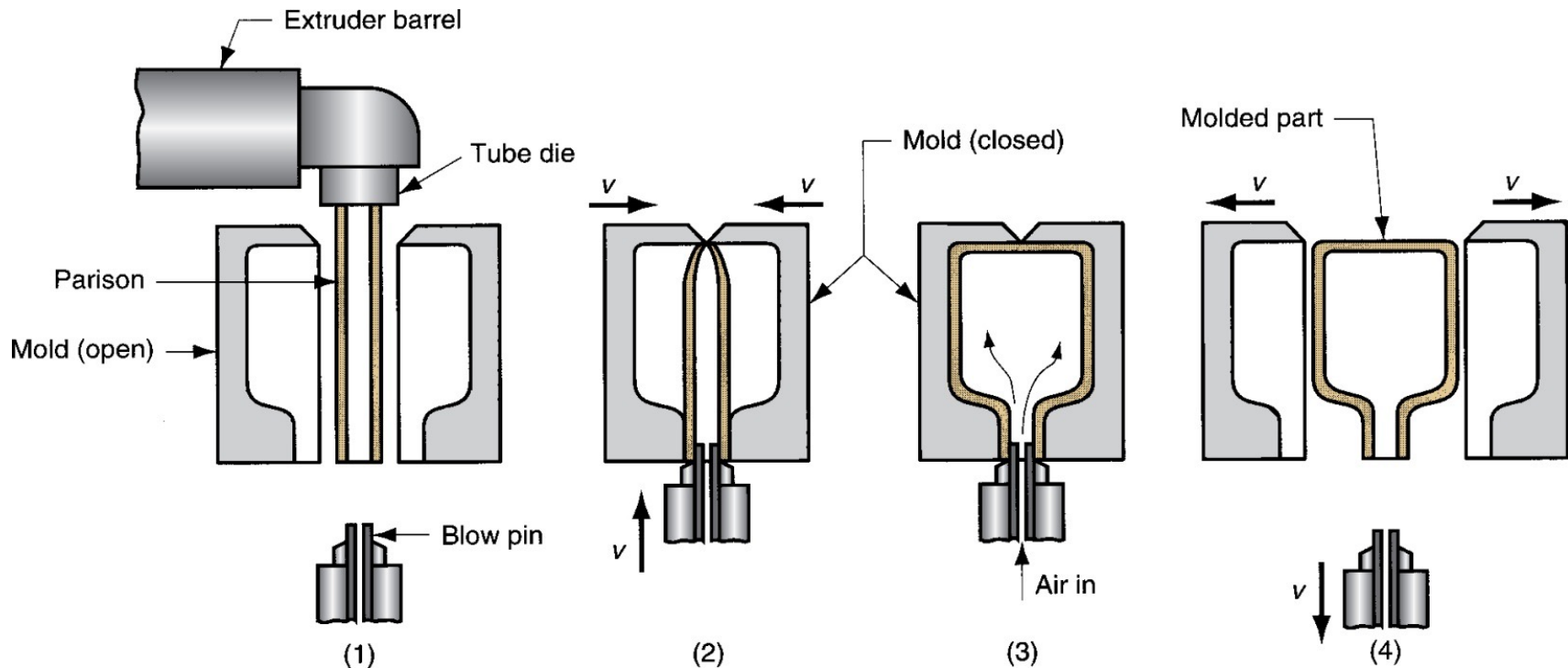
- (1) unload-load,
- (2) heat and rotate mold,
- (3) cool the mold.





EXTRUSION BLOW MOLDING

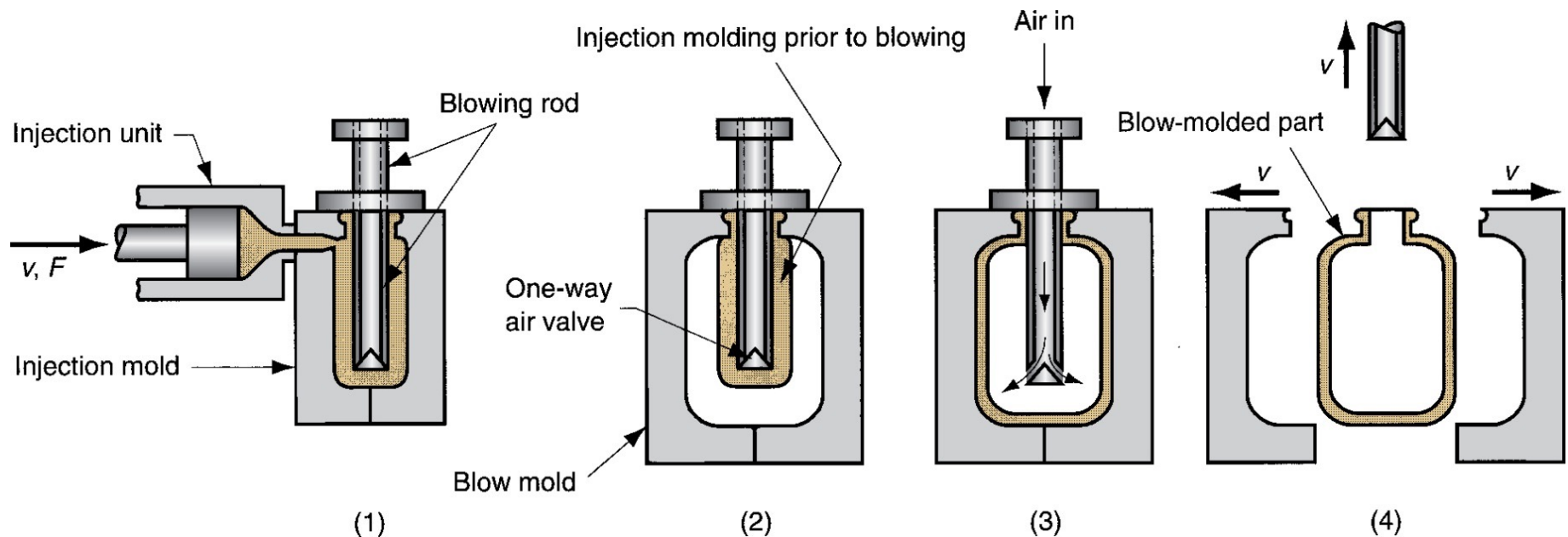
(1) Extrusion of parison; (2) parison is pinched at top and sealed at bottom around blow pin as the mold closes; (3) tube is inflated; and (4) mold is opened.





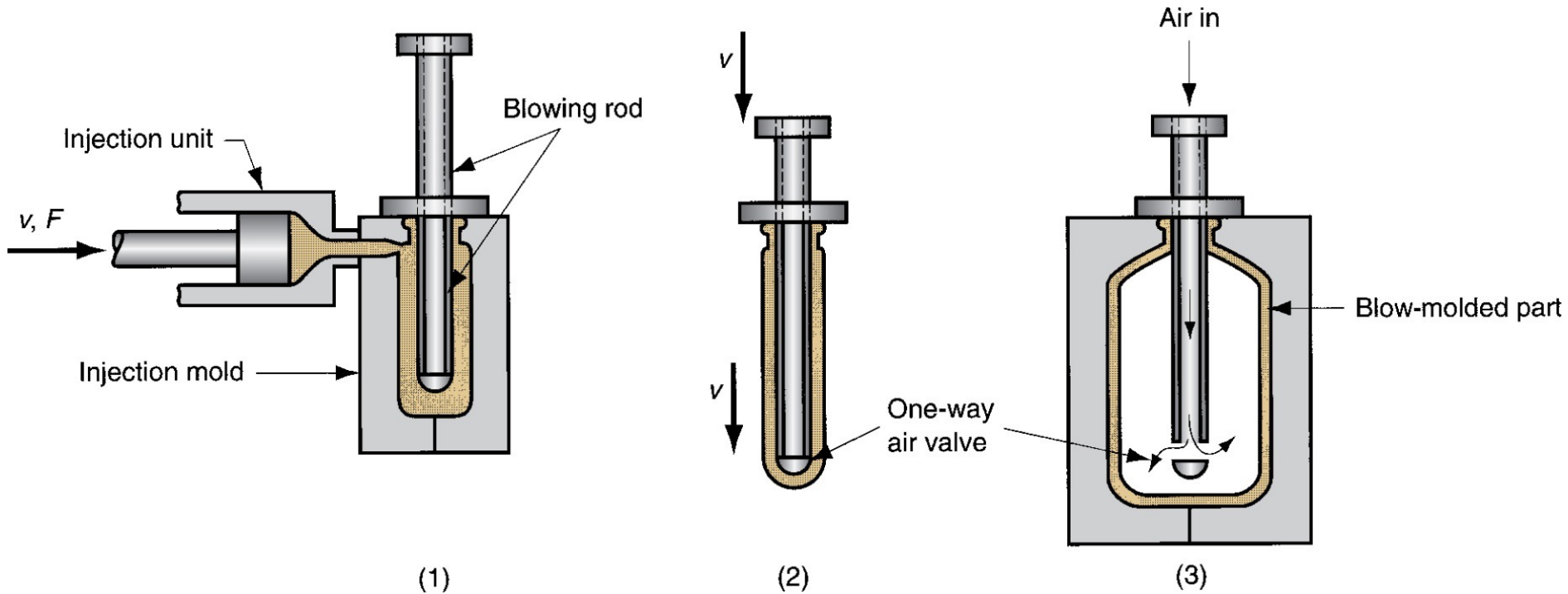
INJECTION BLOW MOLDING

(1) Parison is injected molded around blowing rod; (2) mold is opened, and parison is moved to blow mold; (3) soft polymer is inflated; and (4) mold is opened





(1) Injection molding of parison; (2) stretching; and (3) blowing





MATERIALS AND PRODUCTS IN BLOW MOLDING

Blow molding is limited to *thermoplastics*

Materials: high density polyethylene, polypropylene (PP), polyvinylchloride (PVC), and polyethylene terephthalate

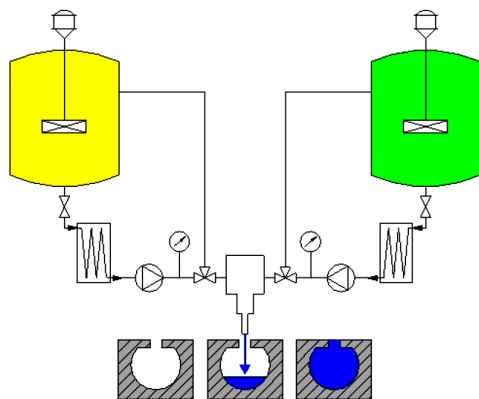
Products: disposable containers for beverages and other liquid consumer goods, large shipping drums for liquids and powders, large storage tanks, gasoline tanks, toys, and hulls for sail boards and small boats.



REACTION INJECTION MOLDING

Two highly reactive liquid ingredients are mixed and immediately injected into a mold cavity where chemical reactions leading to solidification occur.

- RIM was developed with polyurethane to produce large automotive bumpers and fenders
 - RIM polyurethane parts have a foam internal structure surrounded by a dense outer skin
- Other materials: urea-formaldehyde and epoxies





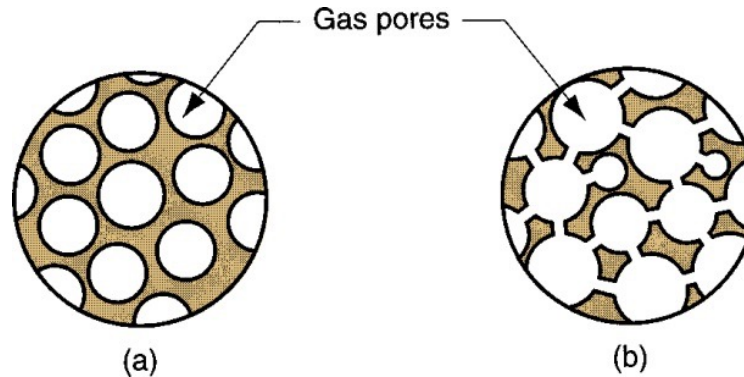
A polymer-and-gas mixture that gives the material a porous or cellular structure:

- Most common polymer foams: polystyrene (Styrofoam, a trademark) and polyurethane
- Other polymers: natural rubber ("foamed rubber") and polyvinylchloride (PVC)

Properties of a Foamed Polymer

- Low density
- High strength per unit weight
- Good thermal insulation
- Good energy absorbing qualities

Two polymer foam structures: (a) closed cell and (b) open cell



- Elastomeric - matrix polymer is a rubber, capable of large elastic deformation
- Flexible - matrix is a highly plasticized polymer such as soft PVC
- Rigid - polymer is a stiff thermoplastic such as polystyrene or a thermoset such as a phenolic
- Applications: heat insulating structural materials, cores for structural panels, packaging materials, cushion materials for furniture and bedding, padding for automobile dashboards, and products requiring buoyancy



Pouring liquid resin into a mold, using gravity to fill cavity, where polymer hardens

- Both thermoplastics and thermosets are cast
 - Thermoplastics: acrylics, polystyrene, polyamides (nylons) and PVC
 - Thermosets: polyurethane, unsaturated polyesters, phenolics, and epoxies
- Simple mold
- Suited to low quantities

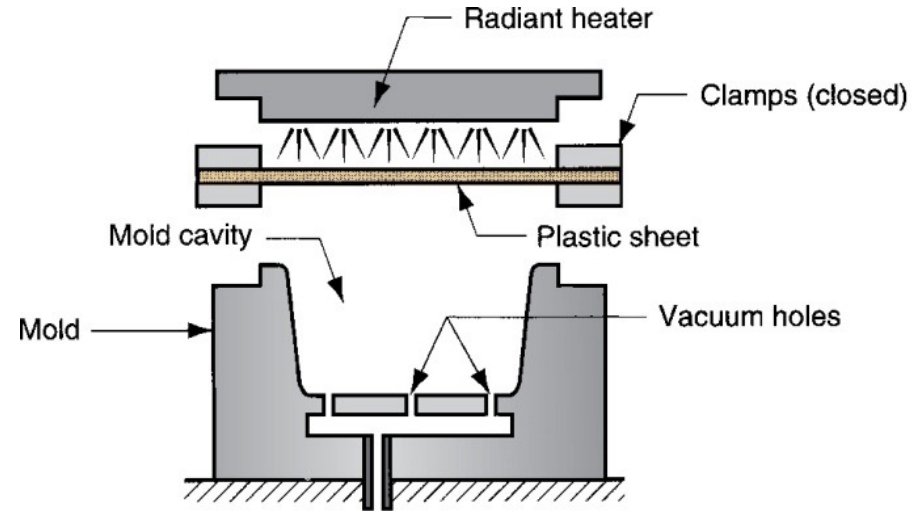


Flat thermoplastic sheet or film is heated and deformed into desired shape using a mold:

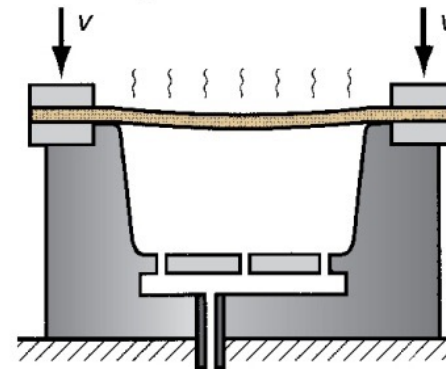
- Heating usually accomplished by radiant electric heaters located on one or both sides of starting plastic sheet or film;
- Widely used in packaging of products and to fabricate large items such as bathtubs, contoured skylights, and internal door liners for refrigerators.



(1) Flat plastic sheet is softened by heating

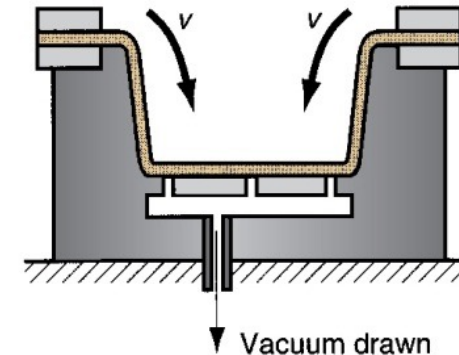


(2) The softened sheet is placed over a concave mold cavity

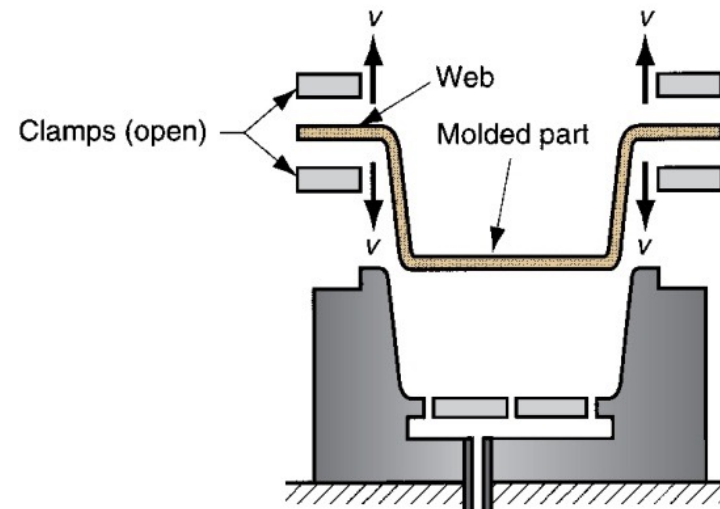




(3) Vacuum draws the sheet into the cavity



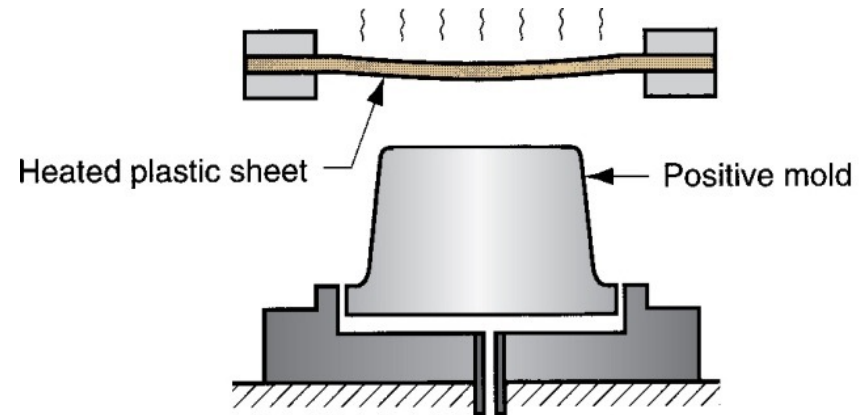
(4) Plastic hardens on contact with the cold mold surface, and the part is removed and later trimmed from the web



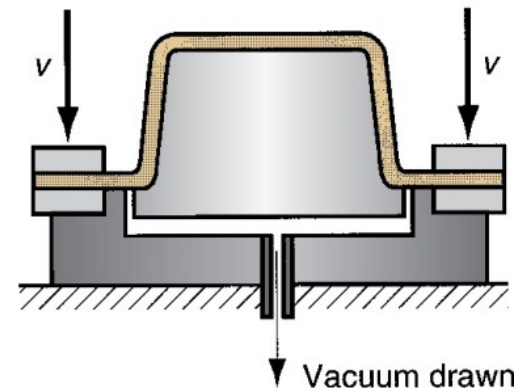


VACUUM THERMOFORMING USING A POSITIVE MOLD

(1) Heated plastic sheet is positioned above the convex mold



(2) Clamp is lowered into position, draping the sheet over the mold as a vacuum forces the sheet against the mold surface



MATERIALS FOR THERMOFORMING

Only thermoplastics can be thermoformed

- Sheets of thermosetting or elastomeric polymers have already been cross-linked and cannot be softened by reheating

Common TP polymers: polystyrene, cellulose acetate, cellulose acetate butyrate, ABS, PVC, acrylic (polymethylmethacrylate - PMMA), polyethylene (PE), and polypropylene (PP).



